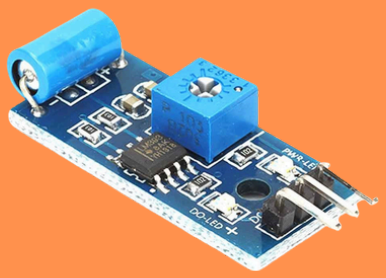
Topic 5

Topic Name: Introduction to Sensors





Just like we use our senses to explore the world, sensors help machines do the same!

What are Sensors?

Sensing Today for a Brighter Tomorrow!

Sensors are like the sense organs of a human being. They help a computer or a microcontroller, like Arduino, to "see", "hear", "feel" and "know" what is happening in the real world.

How Do Sensors Work?

Sensors detect a change in their surroundings (e.g., light, sound, distance, touch, moisture, etc.) and convert it into an electrical signal that the Arduino can understand.

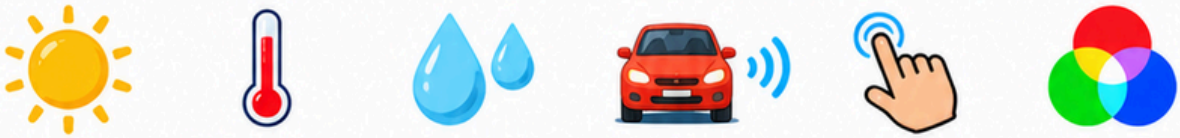


Real World Input → Sensor → Electrical Signal → Arduino Processes the Signal

Why are Sensors Important?

- Help Arduino interact with the real world
- Make projects smart and automatic
- Used in many real-life devices
- Open up endless possibilities for creative projects

Examples of Things Sensors Can Detect



- **Light** (How bright or dark it is)
- **Temperature** (How hot or cold it is)
- **Moisture** (How much water is present)
- **Distance** (How far an object is)
- **Touch** (When something is pressed)
- **Color** (What color an object is)

Sensors are Everywhere!

Sensors are used in many devices around us.



- **Smartphones** (adjust screen brightness)
- **Automatic Doors** (detects motion)
- **Smart Farming** (checks soil moisture)
- **Robot Vacuum** (detects obstacles)



Sensors help machines understand the world, just like our five senses help us!

Small Sensors Big Ideas!



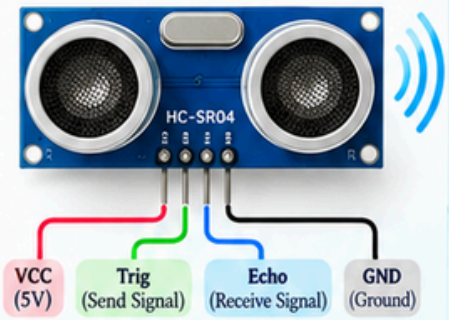
Meet the Sensors

Different Sensors Different Superpowers!

Each sensor can detect something different from the environment.
Let's explore some of the most commonly used sensors!

Ultrasonic Sensor

Measures the distance between the sensor and an object using sound waves.

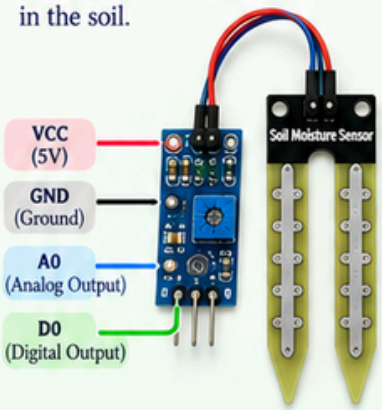




What it detects:
Distance of objects
(e.g., how far an object is).

Soil Moisture Sensor

Checks how much water is present in the soil.

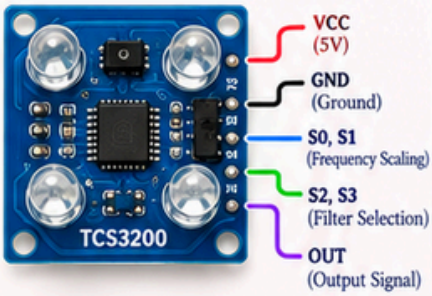




What it detects:
Moisture in soil
(e.g., dry or wet soil).

Color Sensor

Detects the colour of an object by measuring the reflected light.

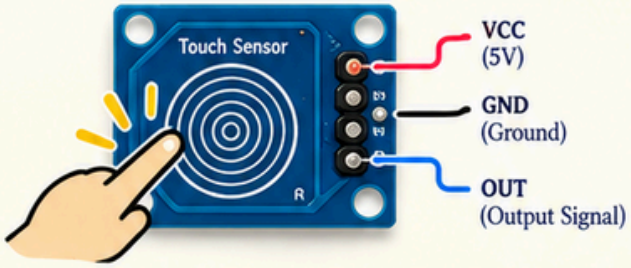




What it detects:
Colour of objects
(e.g., red, blue, green, etc.).

Touch Sensor

Detects when you touch the sensor.





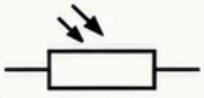
What it detects:
Touch or contact
(e.g., when you touch the sensor).

LDR (Light Dependent Resistor)

Detects the amount of light in the surroundings.



Symbol



In a circuit diagram, LDR is shown like this.



What it detects:
Light intensity
(e.g., bright light or low light).

Different sensors help us see, feel and understand the world around us!





Real-World Applications of Sensors

Same Sensors Brighter World!

Sensors are used in many real-life devices to make our lives easier, safer and smarter. Let's see some interesting examples for each sensor!

Ultrasonic Sensor



Measures the distance between the sensor and an object using sound waves.

Real-World Example



Used in car parking sensors to detect nearby objects and avoid collisions.

Soil Moisture Sensor



Checks how much water is present in the soil.

Real-World Example



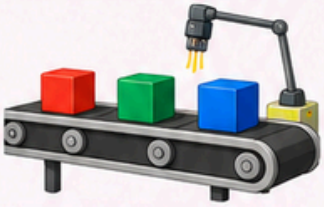
Used in smart irrigation systems to water plants only when the soil is dry.

Color Sensor



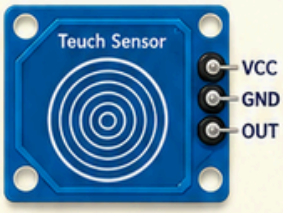
Detects the colour of an object by measuring the reflected light.

Real-World Example



Used in factories to sort objects based on their colour.

Touch Sensor



Detects when you touch the sensor.

Real-World Example



Used in touchlights, water dispensers and many home appliances.

LDR (Light Dependent Resistor)



Detects the amount of light in the surroundings. Its resistance changes based on the light intensity.

Real-World Example



Used in automatic street lights to turn on at night and off during the day.



Sensing Today for a Smarter Tomorrow!

Greener Environment | Safer Communities | Smarter Devices | Brighter Future

Small Sensors Create Big Changes!

Learning Outcomes

- The students will be able to explain what a sensor is and how it helps machines detect changes around them.
- The students will be able to recognise ultrasonic, soil moisture, colour, touch, and LDR sensors and tell what each one detects.
- The students will be able to describe in simple words how sensors collect information and send signals to Arduino.
- The students will be able to connect real-world examples of how sensors make devices smarter and more useful.